EV ChargeSight: Charging Infrastructure Management

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1. Introduction

EV ChargeSight is a web platform that helps a charging-station operator see the health of every charger, turn faults into maintenance tickets, decide which ticket to fix first and send the right technician. It is a working software prototype (MVP) built by our team. Charger behaviour in the demo comes from a built-in **simulator**; there is no live link to a real operator's chargers yet, and the checkout is a **test-mode prototype with no real payment gateway**.

2. Problem statement

India had more than 29,000 public EV charging stations by early 2026 [5], and the PM E-DRIVE scheme set aside ₹2,000 crore for about 72,000 more [6], [7]. More chargers do not help if they are broken: a study of open public DC fast chargers found 22.7 % non-functional [8], and J.D. Power reported that 14 % of U.S. public charging visits in 2025 still failed [9]. Small operators often track faults in spreadsheets and phone calls, so repairs are slow, priorities are unclear and nobody can measure uptime or repair time.

3. Objectives

1. Show the live status of all chargers and sites on one dashboard and map.
2. Convert fault events into tickets automatically, without duplicates, with an SLA timer.
3. Rank tickets by a transparent priority score and assign a qualified technician.
4. Report uptime, MTTA, MTTR and fault trends, with CSV and printable exports.
5. Offer the product as multi-tenant SaaS with roles and subscription plans.

4. Target users

- **Charge point operators (CPOs)** with 5–100 chargers.
- **Fleet depots** (e-bus, e-cab, delivery) that charge their own vehicles.
- **Real-estate owners**: malls, offices, apartments with chargers.
- Inside each customer: owner, admin, operations manager, technician and viewer (the five built-in roles).

5. Purpose, benefits and limitations

Purpose and benefits	Current limitations (stated honestly)
Single place for charger status, tickets and technicians	No live operator integration; charger data in the demo is simulated
Priority formula P = 0.4S + 0.3U + 0.2R + 0.1A ranks work objectively (severity, urgency, revenue impact, age, 24 h cap against starvation)	Checkout is TEST mode only: published test cards and *@test* UPI IDs; no CVV, OTP or PIN is collected and no money moves
SLA timers (4 h critical, 12 h major, 48 h minor; our assumptions) and MTTA/MTTR reports	SQLite database suits a pilot; PostgreSQL is planned before scale
Role-based access, tenant isolation and audit log protect each customer's data	Google Places is not configured; the OpenStreetMap request timed out, so no OSM records are stored
Low running cost: one small cloud server can host a pilot	OCPP 1.6J has been exercised only against our local simulator, not real hardware

6. Technology stack

Layer	Technology used in the repository
Languages	Python 3 (backend), JavaScript ES modules, HTML and CSS (frontend), SQL through an ORM
Backend framework	FastAPI with Pydantic v2 validation; 176 routes, each behind a deny-by-default security guard
Database	SQLite in WAL mode through SQLAlchemy 2.0; 33 tables (organisations, users, sites, chargers, events, tickets, technicians, subscriptions, audit and more)
Frontend	Vanilla JavaScript single-page console at `/app` plus a public website (landing, pricing, sign in, sign up, demo); no heavy framework
Security	Argon2id password hashing, HttpOnly session cookie, CSRF token header, five roles, `org_id` tenant filter, plan entitlements, audit log
Testing	pytest API tests and Playwright real-browser tests: 397 tests, latest full run 397 passed, 0 failed, 0 skipped

7. APIs and how the frontend talks to the backend

The browser never touches the database. Every screen calls a backend REST endpoint with `fetch()` and receives JSON. Signing in sets an **HttpOnly session cookie** that JavaScript cannot read; every state-changing request (POST) must also send the **X-CSRF-Token** header. The guard then checks session → CSRF → role → plan before the request reaches a route, and every database query is filtered by the caller's organisation. Main endpoint groups (all present in the code):

Group	Example endpoints	Purpose
Authentication	`/api/auth/signup`, `signin`, `signout`, `me`, `presentation-demo`	Accounts, sessions, open the demo workspace
Operations	`/api/fleet/summary`, `/api/chargers`, `/api/sim/start`, `/api/sim/step`	Fleet KPIs, charger list, run the simulator
Tickets	`/api/tickets` (GET/POST), `/api/tickets/{code}/assign`, `/api/tickets/{code}/transition`, `/api/priority/ranking`	Create, assign, move and rank tickets
Analytics and reports	`/api/analytics/timeseries`, `/api/timeline`, `/api/reports/{name}.csv`, `/api/reports/full.html`	Charts, event history, CSV and printable reports
Integrations	`/api/integrations/health`, `/api/integrations/operator/test`, `/api/integrations/import/events`, `/api/public-stations`	Source health, CSV import, public station listings
Billing (TEST)	`/api/billing/checkout`, `…/{cid}/payment-method`, `…/{cid}/complete`, `…/{cid}/cancel`	Plan checkout prototype; DEMO receipt

8. External integrations: implemented vs planned

Integration	Status	What it really does today
Open Charge Map API	CONNECTED	Public charging-station listings; a snapshot of 163 listings is stored
Google Maps JavaScript API	CONNECTED	Draws the map in the browser (optional key; verified in a browser)
CSV import	CONNECTED	Operator exports can be uploaded as events
OpenStreetMap Overpass	ERROR	Last request timed out (ConnectTimeout); no OSM records stored
Google Places API	NOT CONNECTED	Code path exists; no key configured and no paid calls made
Operator REST API	NOT CONNECTED	Connection test endpoint only; no real operator linked
OCPP 1.6J	SIMULATED	Tested only against the local simulator
Payment gateway	PLANNED	Not integrated; checkout is a TEST-mode prototype

9. System architecture and data flow

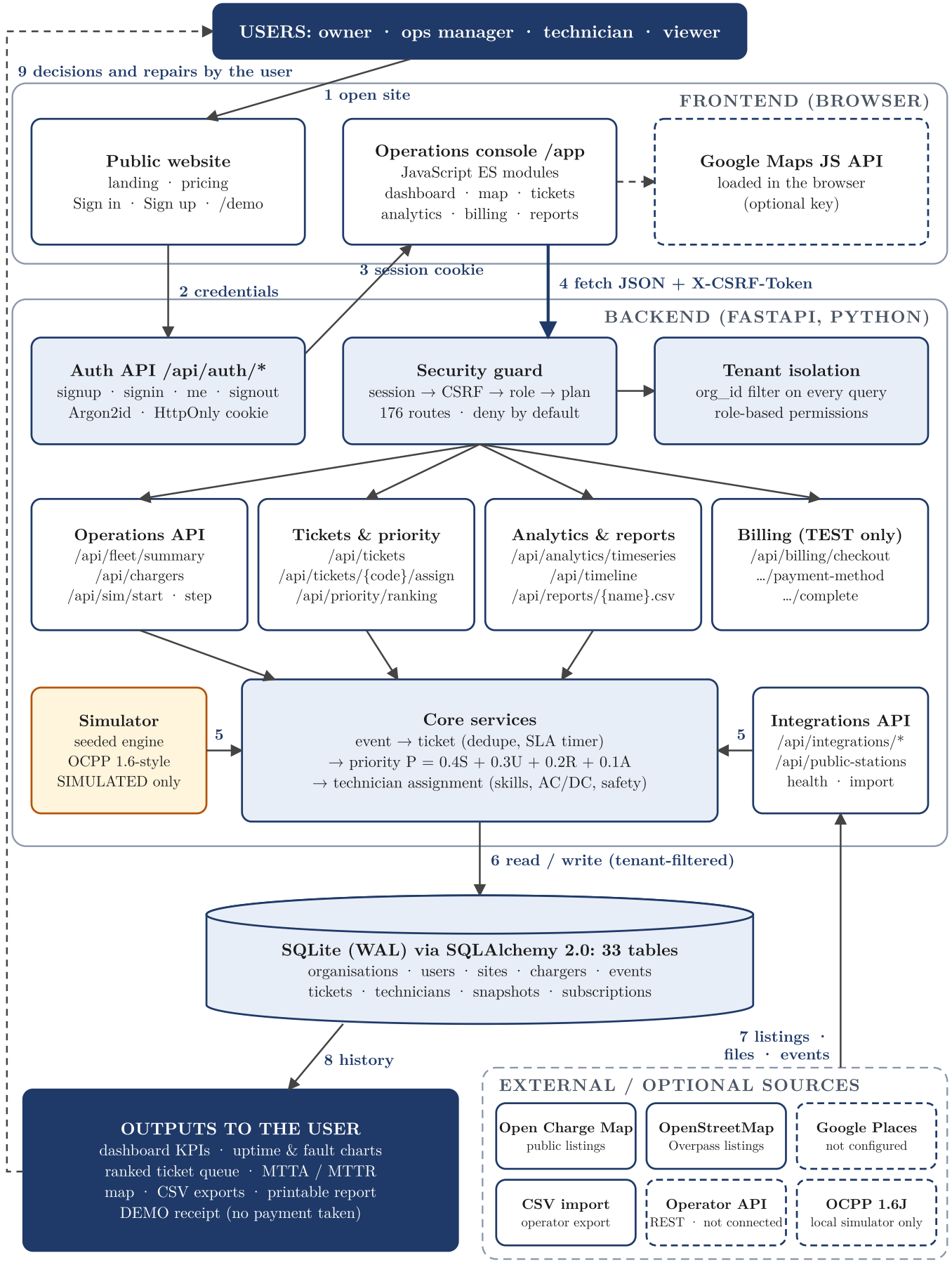


Figure 1. End-to-end architecture. Solid boxes are implemented; dashed boxes are optional, not configured or not connected. Numbers 1–9 follow the data flow explained in Section 10.

10. Data flow from start to finish

1. **Open site:** a user opens the public website and chooses Sign in, Sign up or Start free demo.
2. **Credentials:** the form posts to `/api/auth/signin` (or `signup`); the password is checked with Argon2id.
3. **Session cookie:** the server sets the HttpOnly cookie and redirects to the console `/app`. Without a session, `/app` redirects to sign-in.
4. **API calls:** the console fetches JSON with the cookie and the X-CSRF-Token header; the guard checks role and plan, and the tenant filter limits data to the user's organisation.
5. **Events:** the simulator (seed 42, 15-second steps) or an import produces charger events. Core services deduplicate them, open tickets with SLA timers, compute the priority score and suggest a technician by skills (AC/DC) and safety rules.
6. **Store:** everything is written to and read from the SQLite database (33 tables), always filtered by `org_id`.
7. **External data:** Open Charge Map listings and CSV files enter through the Integrations API; unconnected sources only show their status.
8. **History to outputs:** analytics read the stored history to draw uptime and fault charts, the ranked queue, MTTA/MTTR, the map, CSV exports and the printable report.
9. **Decisions:** the manager assigns and closes tickets in the console; each change is audited and the loop repeats.

11. Prototype (MVP) and how to demonstrate it

MVP features: public website and plans; sign up and sign in; private Presentation Demo workspace; fleet dashboard and map; ticket queue with priority ranking, assignment and status transitions; technician roster; analytics charts; CSV and printable reports; integration health page; TEST-mode plan checkout with a DEMO receipt; experiments and site-investment calculator.

Demo data: 6 sites, 40 simulated chargers, 4 fictional technicians, 6 hours of history and 8 scripted tickets.

Demonstration script (about 8 minutes):

1. Open the website → *Start free demo* → sign up.
2. Show the dashboard KPIs and the map.
3. Run simulator steps; a new fault becomes a ticket.
4. Open the queue: explain the priority score and SLA timer.
5. Assign a technician; move the ticket to resolved.
6. Show analytics (uptime, MTTR) and download a CSV report.
7. Show Integrations: what is connected and what is not.
8. Run a TEST checkout and show the DEMO receipt.

Simulated versus genuinely connected

Component	Status	Note
Charger status, faults, sessions in the demo	SIMULATED	Generated by the seeded simulator; clearly labelled in the UI
Technicians and tickets in the demo	SIMULATED	Fictional names and scripted tickets
Accounts, roles, tenancy, tickets you create, audit log	CONNECTED	Real application logic and real database rows
Public station listings (Open Charge Map)	CONNECTED	Real public data (stored snapshot)
Map display (Google Maps JS)	CONNECTED	Real map tiles in the browser
Payments	SIMULATED	TEST cards/UPI only; DEMO receipt; no gateway, no money
Operator chargers (REST/OCPP)	NOT CONNECTED	No live operator integration exists yet

12. Subscription plans and pricing

Prices below are the values configured in the application. Longer billing periods get 5 % (6 months) or 10 % (12 months) off, so Starter costs ₹2,844.30 for 6 months and ₹5,389.20 for 12 months.

Plan	Price / month	Chargers	Users	Sites	Target customer
Demo	₹0	Simulated only	2	–	Evaluation, presentations
Starter	₹499	5	2	1	Small site owner
Growth	₹1,499	25	5	3	Growing CPO; up to 3 integrations
Professional	₹4,999	100	15	10	Regional CPO or fleet depot; up to 10 integrations
Enterprise	Negotiated	Custom	Custom	Custom	Large networks

Revenue streams

(1) Monthly or yearly subscriptions (main stream); (2) one-time onboarding and data-import service; (3) Enterprise contracts with custom integrations; (4) later, paid add-ons such as extra integrations or SMS alerts. Only stream (1) is modelled below.

13. Operating costs (monthly)

Item	INR / month	Basis
Production server: Basic Droplet 2 GiB / 1 vCPU (US$12)	₹1,056	DigitalOcean price list [10]
Weekly backups (20 % of Droplet price)	₹211	DigitalOcean price list [10]
Staging server: Basic Droplet 1 GiB (US$6)	₹528	DigitalOcean price list [10]
Maps: Google Maps within free monthly caps	₹0	Free usage per SKU [11]; OpenStreetMap fallback
Open Charge Map public listings	₹0	Free API key (registered)
Transactional e-mail (password reset, invitations)	₹500	ASSUMPTION: entry tier
Domain amortised (1,000 / 12)	₹83	From one-time cost
Maintenance and support effort (20 h)	₹5,000	Notional effort at the assumed rate
Marketing: LinkedIn ads, one 14-day burst per quarter at the US$10/day minimum [13] (₹12,320/quarter)	₹4,107	ASSUMPTION
Marketing: events, demos, brochures	₹2,000	ASSUMPTION
Total operating cost per month	**₹13,485**	ESTIMATE
Total per year (× 12)	**₹1,61,816**	ESTIMATE

USD converted at ₹88 per US$ (assumption). Cash costs excluding marketing are ₹2,378/month; support effort is notional.

14. Break-even example ESTIMATE

Assumed customer mix 60 % Starter, 30 % Growth, 10 % Professional gives an average revenue per customer (ARPU) of $0.6 \times 499 + 0.3 \times 1{,}499 + 0.1 \times 4{,}999 =$ **₹1,249** per month, or **₹1,224.02** after an assumed 2 % gateway fee.

Scenario	Monthly cost	Customers needed (cost ÷ net ARPU, rounded up)
Cash costs only (servers, e-mail, domain)	₹2,378	2,378 ÷ 1,224.02 = 1.94 → **2**
Plus notional support effort	₹7,378	7,378 ÷ 1,224.02 = 6.03 → **7**
Plus marketing (full operating cost)	₹13,485	13,485 ÷ 1,224.02 = 11.02 → **12**

Customer acquisition cost (CAC), assuming 2 new customers a month: ₹6,107 ÷ 2 ≈ **₹3,053**, recovered in about 2.5 months of net ARPU. All forecasts are hypotheses; there are no paying customers today.

15. Entrepreneurial pitch

Problem	Broken chargers and slow, unmeasured repairs hurt drivers and operator revenue [8], [9].
Solution	A maintenance-first SaaS: fault → ticket → priority → technician → report, in one console.
Unique value	Transparent priority formula with SLA and starvation cap, technician matching by skill and safety, India-friendly pricing from ₹499/month, honest data labels (simulated vs connected).
Market	29,000+ public stations in India [5] and ~72,000 more planned under PM E-DRIVE [6], [7]; first segment: small and mid-size CPOs and fleet depots.
Competitors	AMPECO [14] and Driivz [15] are full charging-management platforms for large networks; Kazam [16] combines hardware and software. ChargeSight focuses narrowly on maintenance operations for smaller operators.
Business model	Tiered monthly subscription (Starter, Growth, Professional, Enterprise) plus onboarding services.
Growth strategy	Free demo → pilot with one operator → case study → LinkedIn and EV-industry events → partnerships with charger installers.
Future scope	Live OCPP and operator API links, a real payment gateway, PostgreSQL, a mobile technician app, predictive fault alerts.

16. PME concepts applied

16.1 Feasibility (pre-feasibility and feasibility study)

Type	Finding
Technical	Feasible: MVP works end to end with 397 passing tests; open standards (OCPP, REST, CSV) exist for real integration.
Economic / financial	Low cost: ₹2,378/month cash cost for a pilot; break-even at 7–12 customers (estimate).
Market / commercial	Growing station count and government support [5]–[7]; reliability is a known industry problem [8], [9].
Operational	Four-person team; simple web UI; runs on one small server.
Legal	Needs terms, privacy policy and data-processing agreement before a real pilot (budgeted).

16.2 Work Breakdown Structure (summary)

WBS	Phase	Main work packages (activity IDs)
1	Initiation and analysis	Charter (A), requirements (B), research (C), feasibility (D)
2	Design and development	Architecture (E), backend and simulator (F), console (G), tickets and priority (H), analytics (I), integrations (J), SaaS layer (K), UI integration (L)
3	Testing	Testing and validation (M)
4	Commercial and closure	Business model and pricing (N), documentation (O), final presentation (P)

16.3 CPM and PERT results

Sixteen activities (A–P) were networked with their dependencies. The forward and backward pass give a project duration of **T = 47 working days** on the critical path **A–C–D–E–F–H–L–M–O–P** (zero total float). PERT uses te = (o + 4m + p) / 6: the critical path has expected time **48.5 days** and standard deviation **σ** = 1.96 days, so the probability of finishing within 50 days is Z = (50 − 48.5) / 1.96 = 0.76 → **77.7 %**. Crashing was checked for F, H and M (cost slopes H ₹5,500/day, F ₹6,000/day, M ₹8,000/day); every slope is higher than the assumed indirect saving of ₹3,000/day, so the normal schedule is the cheapest.

16.4 Gantt chart

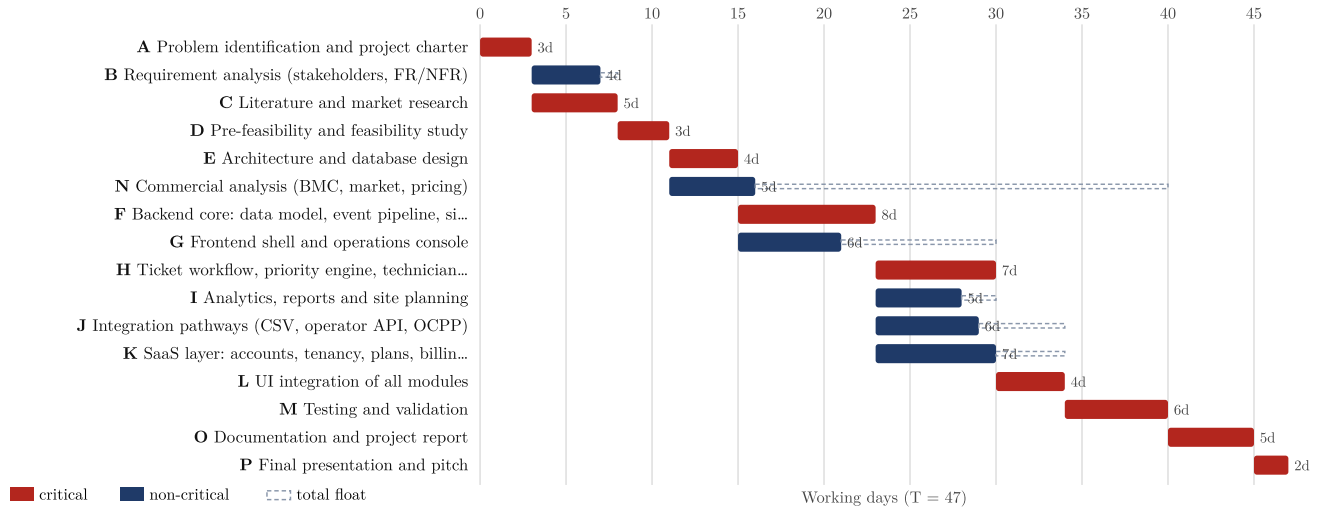


Figure 2. Planned schedule of activities A–P in working days; critical activities in red, float shown dashed.

16.5 Resource planning

Role	Main responsibility
Project manager	Charter, feasibility, commercial analysis, report, pitch
Backend lead	Architecture, API, simulator, tickets, integrations, SaaS layer
Frontend / UX lead	Console, analytics screens, UI integration
Business analyst / QA	Requirements, research, testing

Effort: 4 members × 47 days × 4 h = 752 person-hours; at a notional ₹250/h this is ₹1,88,000 (not a cash cost).

16.6 Budget (first year) ESTIMATE

Item	INR
One-time (domain, security review, legal, pilot onboarding, DB migration)	₹76,000
Recurring: 12 × ₹7,378	₹88,536
Contingency 10 %	₹16,454
First-year budget (excl. marketing)	**₹1,80,990**

Hypothetical 3-year NPV at 12 % (10/25/45 customers at year ends): ₹3,28,094.

16.7 Risk management

Risk	Probability / impact	Response
No operator agrees to a pilot	Medium / High	Free demo, small paid pilot, case study; target fleet depots too
Real charger data differs from the simulator	High / Medium	CSV import first; OCPP test with real hardware before launch
Security or data-leak incident	Low / High	Tenant filter, CSRF, Argon2id, audit log; external security review budgeted
Schedule slip on critical path	Medium / Medium	Monitor critical activities; 78 % chance of ≤ 50 days; crash H first if needed

16.8 Business Model Canvas

Customer segments Small/mid CPOs, fleet depots, real-estate owners with chargers	**Value proposition** Fewer broken chargers, faster repairs, clear priorities, measurable uptime	**Channels** Website and free demo, LinkedIn, EV events, installer partners
Customer relationships Self-service plans, onboarding help, Enterprise account management	**Revenue streams** Subscriptions (₹499 / ₹1,499 / ₹4,999 per month), onboarding, Enterprise contracts	**Key resources** Software platform, priority engine, team skills, data
Key activities Development, integrations, support, sales	**Key partners** Charger installers, Open Charge Map, cloud host, future payment gateway	**Cost structure** Hosting and e-mail (₹2,378/month), support effort, marketing, security and legal

17. Likely presentation and viva questions

Q1. What problem does ChargeSight solve?
It turns charger faults into prioritised, assigned maintenance tickets so operators repair chargers faster and can measure uptime and repair time.

Q2. Is the charger data real?
No. The demo uses a seeded simulator; only accounts, tickets you create, Open Charge Map listings and the map are real. No live operator integration exists yet.

Q3. Can customers pay through the app?
Not yet. Checkout is a TEST-mode prototype with test cards and @test UPI IDs; no real payment gateway is connected and no money moves.

Q4. How is ticket priority calculated?
$P = 0.4 \cdot \text{Severity} + 0.3 \cdot \text{Urgency} + 0.2 \cdot \text{Revenue impact} + 0.1 \cdot \text{Age}$, with a 24-hour cap so old low-priority tickets are not starved.

Q5. How does the frontend talk to the backend?
Through REST calls returning JSON; the HttpOnly session cookie proves identity and the X-CSRF-Token header protects POST requests.

Q6. How do you keep customers' data separate?
Every query is filtered by the organisation ID, and a deny-by-default guard checks session, CSRF, role and plan on every route.

Q7. What is the break-even point?
About 7 customers without marketing and 12 with marketing, using an assumed ARPU of ₹1,224; these are estimates.

Q8. Who are your competitors?
AMPECO, Driivz and Kazam; we target smaller operators with a maintenance-first, low-price product.

Q9. What is a project? (Unit 1)
A temporary effort with a defined start, end and unique result; it moves through initiation, planning, execution and closure.

Q10. Why do a feasibility study? (Unit 2)
To check technical, economic, commercial, operational and legal viability before committing money and time.

Q11. What is a WBS? (Unit 2)
A hierarchical breakdown of the total project scope into smaller deliverables and work packages that can be estimated and assigned.

Q12. What is the critical path? (Unit 3)
The longest path through the network with zero float; ours is A–C–D–E–F–H–L–M–O–P and lasts 47 days.

Q13. Difference between CPM and PERT? (Unit 3)
CPM uses one fixed time per activity and supports time–cost trade-offs; PERT uses three time estimates to handle uncertainty.

Q14. What is crashing? (Unit 3)
Reducing an activity's duration by adding resources; it is worthwhile only when the crash cost slope is below the indirect cost saved.

Q15. What is entrepreneurship and innovation? (Unit 4)
Entrepreneurship is turning an opportunity into a venture while bearing risk; innovation is applying a new idea that creates value.

Q16. Divergent vs convergent thinking? (Unit 4)
Divergent thinking generates many ideas; convergent thinking selects and refines the best one.

Q17. What is a competitive advantage? (Unit 5)
Something that lets a firm serve customers better or cheaper than rivals; ours is a focused, low-cost maintenance workflow.

Q18. What is design thinking? (Unit 5)
A user-centred process: empathise, define, ideate, prototype and test; our demo and pilot feedback follow it.

Q19. What does a Business Model Canvas show?
Nine blocks that describe how a business creates, delivers and captures value, from customer segments to cost structure.

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